

The Schrödinger Equation: From a String's Wave to a Particle's Wave

First-Principles Derivation Series · 05

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The Schrödinger Equation: From a String's Wave to a Particle's Wave

Author's note — on the structure of this post

This Post 05 is unlike the earlier ones: **its weight is in the prerequisites, not the main text.** The two genuinely hard things — where Planck's constant h comes from, and "by what right is a particle a wave" — are both handled separately (h is squeezed out in Part 2, *Squeezing Planck's Constant Out of Blackbody Radiation*; the de Broglie seam is faced on its own in §2 here). With the prerequisites carried off, **the main derivation is startlingly short: the core is one step — translate the dispersion relation into an equation via "differentiating a plane wave," a few lines of algebra and it's done.**

This isn't cutting corners; it's dictated by the **nature** of this equation. The Schrödinger equation is a **postulate-level fundamental law** — it is not deduced from any deeper mechanics (below we'll see why it can't be), it can only be "pinned into shape by the properties it must satisfy." So its difficulty is in "why it may be set up this way," not in "the few lines of algebra once set up."

Same rules as before: one side drives the derivation, the other only judges "does it close." Where things got stuck and how they got unstuck I keep as intact as I can, marked Checkpoint. Where it can't be derived and can only be accepted as a postulate, I record an **IOU**, collected at the end — first principles isn't welding shut every foundation slab, it's **knowing why each step holds and honestly marking which pieces are pegs.**

0. Where this block sits, and its relation to 04

- **Backward:** 04 derived the 1D string's wave equation, handing us a **whole “language of waves”** — plane waves $e^{i(kx-\omega t)}$, the dispersion relation $\omega(k)$, the “equation $\leftrightarrow$ dispersion” machine, mode decomposition, $\Delta x \cdot \Delta k \geq \frac{1}{2}$. This post transplants that language **onto a “particle.”**
- **Forward:** the Schrödinger equation is the dynamical core of all quantum mechanics; 06 (states, operators, measurement) upgrades the “wave function” appearing here into a Hilbert-space state vector; 08 (relativistic quantum equations) will settle the IOU left here by using non-relativistic energy.

This post lands on one equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\mathbf{r})\psi$$

But more important than the equation is **by what right it looks like this.**

1. What this equation describes (first fix the object being modeled)

Before deriving, ask: what is being described? Otherwise it becomes “knowing the answer and faking the steps.”

The object described is the wave function $\psi(x, t)$. Not the particle's position, not a trajectory (a quantum particle has no definite trajectory), but a **complex-valued function spread over space**. Its physical meaning is the Born rule (a postulate accepted in 03):

$$|\psi(x, t)|^2 = \text{the probability density of finding the particle near position } x \text{ at time } t$$

The Schrödinger equation answers: **given $\psi(x, 0)$ at some instant, how does it change with time** — i.e. the **evolution law** for ψ . In status it parallels Newton's second law:

	Object described	State	Evolution equation
Classical	one point mass	(x, p) : position + momentum	$F = ma$ (second order)
Quantum	one particle	the whole function $\psi(x)$	Schrödinger (first order in time)

Checkpoint 1: what evolves is a “probability distribution,” not “the next landing point”

“Predict where the particle will be next” is the wrong question — the landing point of a single measurement is random and unpredictable. What is predictable and evolvable is the **whole wave function ψ** (and the distribution $|\psi|^2$ it gives). So the Schrödinger equation doesn't predict “the next position,” it evolves “the next instant's probability amplitude.” Landing point random, distribution determinate.

Checkpoint 2: Δx is “the standard deviation of the distribution you get by treating $|\psi|^2$ as a probability density,” not a subtraction of two points, not the string length

On a 1D string, the string’s physical width is 0 to L , fixed — that is **not** Δx . Δx arises like this: treat $|\psi(x)|^2$ as a probability density over x (normalized, $\int |\psi|^2 dx = 1$); it forms a probability distribution, and Δx is the **standard deviation of that distribution**:

$$\langle x \rangle = \int x |\psi|^2 dx, \quad \Delta x = \sqrt{\int (x - \langle x \rangle)^2 |\psi|^2 dx}$$

It is not “ $x_2 - x_1$,” two specific positions subtracted (a wave packet’s tails are gradual, with no clean endpoints to pick out); it is “how wide this $|\psi|^2$ spreads, by weight.” Numerically it’s about the width your eye sees, but the definition must be the standard deviation to be unique and computable. — This anchor was asked about repeatedly and finally nailed: $\Delta x = \text{std of the distribution } |\psi|^2$.

1.1 Why it’s “first order” in time: the state is different

To **fully fix a classical point mass’s present state** and predict the future, you need two quantities: position x and momentum p . Because Newton’s second law is **second order** ($F = m\ddot{x}$), a second-order equation needs **two** initial conditions (initial position + initial velocity) for a unique solution. So the classical state is (x, p) — in 3D, $(\vec{x}, \vec{p})$, six components, with the direction of motion encoded in the orientation of the momentum vector $\vec{p}$.

Quantum is different: **the complete state is the whole function $\psi(x)$ itself** (it already contains all the information about the position distribution). So the evolution equation needs only a **first-order** time derivative — given $\psi(x, 0)$, this **one** function, you can fix $\psi(x, t)$ thereafter; you don’t need “the time derivative of ψ ” as a second initial condition. This is the root reason the Schrödinger equation is first order in t (whereas 04’s wave equation is second order in t).

1.2 One particle, not a swarm

$\psi(x, t)$ describes **a single particle**; $|\psi|^2$ is the probability density for finding **that one** particle. “Repeat many times to get a distribution” means **identical preparation, independent repeated measurement** — each run is “prepare one particle $\rightarrow$ measure once,” with landing points following $|\psi|^2$. It is **one particle’s positional uncertainty**, not “many particles spread out.”

Multiple particles isn’t “several equations added”: the wave function of N particles is $\psi(x_1, \dots, x_N, t)$ — **one** function on an N -dimensional configuration space, still **one** Schrödinger equation. (Dimension grows with particle number — the root of why quantum many-body is hard.) This post does the **single particle** throughout, which is where quantum mechanics actually begins (Schrödinger’s 1926 solution of hydrogen was a single electron), not a simplification into falsehood.

2. The fork from 04 to 05: why you can’t reuse “force analysis”

This is the step most worth making clear — it dictates that 05’s derivation runs **opposite** to 04’s.

How 04 was driven: there was a **real medium** (the string) + Newton's laws. Do a **force analysis** $F = ma$ on a small segment: force (tension), mass (μdx), acceleration are all physical realities, write Newton's equation directly $\rightarrow$ the wave equation. **The equation was derived from the underlying mechanics.**

Checkpoint 3: 05 can't reuse this, because the bottom is cut off

The 1D particle's $\psi(x, t)$ has **no medium, no thing being acted upon**. ψ is a probability amplitude, not the displacement of some matter — you **cannot do** $F = ma$ on it: there's no "force" acting on a probability amplitude, no "mass element." So 04's route "force analysis $\rightarrow$ equation" is **cut off** in 05.

This isn't a technical difficulty, it's a **level** issue: a particle's matter wave is a **new-level fundamental law** — there's no more basic mechanics beneath it from which to derive it. So the modeling method must switch:

- **04: have an equation** (Newton gives it) $\rightarrow$ substitute a plane wave $\rightarrow$ **obtain** the dispersion relation $\omega = ck$.
- **05: no underlying equation to derive from** $\rightarrow$ can only **require** "this equation must make plane waves satisfy the correct ω - k relation" $\rightarrow$ **reverse-engineer** the equation's form.

The direction is reversed. This is **not reverse-engineering-from-the-answer** — at the "fundamental law" level, the equation itself is postulate-level, not deduced from below, and can only be pinned into shape by "what it must satisfy." Every fundamental equation in physics is set up this way.

2.1 The seam: by what right describe a particle with a wave

Before "describing a particle with a wave ψ ," there's no place yet for momentum, energy, let alone which energy relation. What must be crossed first is this seam itself:

Checkpoint 4: "a particle is a wave" is a postulate, not a derivation

Electrons diffract, they interfere — they behave like waves. But "a particle has a wave ψ " **cannot be derived**; it is a **physical assumption / experimental fact**. Only after crossing this seam — granting "a particle can be described by a wave ψ " — does it make sense to ask "how do this wave's ω, k connect to the particle's E, p ."

After crossing, what connects "wave properties" to "particle mechanical quantities" is the de Broglie postulate (this post's physical input, accepted like the Born rule, not derived):

$$p = \hbar k, \quad E = \hbar \omega$$

2.2 de Broglie only ferries; it doesn't give the E - p relation

Checkpoint 5: de Broglie alone isn't enough — you still lack the "energy-momentum relation," a third piece

$p = \hbar k$ and $E = \hbar \omega$ merely **translate** the particle's (E, p) into the wave's (ω, k) ; each is one translation, and **they don't tell you the relation between E and p** . To get the relation between ω and

k (the dispersion), you must know how this particle's E and p are linked — an **independent third input**, decided by “what kind of particle this is”:

- non-relativistic free particle: $E = \frac{p^2}{2m} \rightarrow$ gives the Schrödinger equation
- relativistic: $E^2 = (pc)^2 + (mc^2)^2 \rightarrow$ gives Klein–Gordon / Dirac (Ch. 08)
- photon: $E = pc$

Which you pick decides which wave equation you get. This post picks the **non-relativistic free particle** $E = \frac{p^2}{2m}$.

Where $\frac{p^2}{2m}$ comes from (in Part 1's classical mechanics): kinetic energy $E_k = \frac{1}{2}mv^2$, momentum $p = mv$ so $v = p/m$, substitute to get $E_k = \frac{1}{2}m(p/m)^2 = \frac{p^2}{2m}$; “free” = no potential, all energy is kinetic.

So the three inputs are assembled: (i) a particle can be described by a wave ψ (the seam, a postulate); (ii) de Broglie $p = \hbar k$, $E = \hbar\omega$; (iii) the energy–momentum relation $E = \frac{p^2}{2m}$. Together they give the dispersion relation:

$$\hbar\omega = \frac{(\hbar k)^2}{2m} = \frac{\hbar^2 k^2}{2m} \implies \omega = \frac{\hbar k^2}{2m}$$

Note how this dispersion differs from the string's: the string is $\omega = ck$ (linear, first power); the free quantum particle is $\omega \propto k^2$ (second power). Physically, the string has “all k at one phase speed c ” (the packet doesn't spread), the quantum particle has “phase speed $\propto k$ ” (different k part ways, the packet disperses). This is the concrete meaning of 04's repeated line “swap in a different dispersion relation.”

3. The modeling constraints: what the equation must satisfy

With the dispersion in hand, we still must fix the equation's **form**. Three constraints (the first two are properties a “sensible quantum evolution equation” should have; the third connects it to the dispersion):

1. **First-order time evolution:** given $\psi(x, 0)$, $\psi(x, t)$ is uniquely fixed. So the time derivative is **first order** $\partial_t \psi$ (reason in §1.1: the quantum state is ψ itself, first order suffices).
2. **Linearity:** quantum states superpose (the superposition principle, a basic postulate), so waves can interfere. So the equation is linear — if ψ_1, ψ_2 are solutions, so is $a\psi_1 + b\psi_2$.
3. **Plane waves are its solutions, satisfying the correct dispersion:** a free particle's matter wave is a plane wave $e^{i(kx - \omega t)}$, which must be a solution of this equation, and whose ω, k satisfy $\omega = \frac{\hbar k^2}{2m}$.

Checkpoint 6: plane waves are solutions — this is the lever for reverse-engineering the equation

Not “have an equation first, then solve out plane waves,” but “**knowing the solution is a plane wave, constrain the equation's form backward.**” And plane waves are extremely friendly to differentiation (they are eigenfunctions of every ∂_x, ∂_t ; differentiating just multiplies by a constant), which lets “substitute $\rightarrow$ fix the form” actually be computed. This is the handle for the next step.

A note on the basis: 04's string uses $\sin(k_n x)$, 05's free particle uses e^{ikx} — **not a switch to something new**. $\sin, \cos, e^{ikx}$ are all eigenfunctions of the same operator $\frac{d^2}{dx^2}$, the same family of bases picked differently under different **boundary conditions**: 04's fixed ends ($\psi(0) = \psi(L) = 0$) kill $\cos$, leave only $\sin$, and discretize k ; 05's free unbounded case has continuous k , keeps both $\cos + i \sin$, combined into e^{ikx} . Choosing e^{ikx} is also because it's cleanest to differentiate ($\partial_x e^{ikx} = ik e^{ikx}$, one step pulls down ik ; $\sin$ would bounce $\sin \rightarrow \cos \rightarrow -\sin$).

4. The main derivation: differentiate a plane wave $\rightarrow$ substitute $\rightarrow$ equation

The core mechanism in one line: **differentiating a plane wave pulls ω, k down as constants; so you can “substitute away” ω, k with derivatives, translating the dispersion relation into an equation.**

Take the plane wave $\psi = e^{i(kx - \omega t)}$ and differentiate:

$$\partial_t \psi = -i\omega \psi, \quad \partial_x^2 \psi = -k^2 \psi$$

Solve back for $\omega\psi$ and $k^2\psi$:

$$\omega\psi = i \partial_t \psi, \quad k^2\psi = -\partial_x^2 \psi$$

Multiply the dispersion relation by ψ (here a **genuine multiplication**: for a single plane wave, $\hbar\omega$ is a definite number):

$$\hbar\omega \psi = \frac{\hbar^2 k^2}{2m} \psi$$

Replace $\omega\psi$ and $k^2\psi$ with derivatives:

- LHS: $\hbar\omega\psi = \hbar \cdot i \partial_t \psi = i\hbar \partial_t \psi$
- RHS: $\frac{\hbar^2}{2m} k^2\psi = \frac{\hbar^2}{2m} (-\partial_x^2 \psi) = -\frac{\hbar^2}{2m} \partial_x^2 \psi$

giving the **1D free-particle Schrödinger equation**:

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}}$$

Checkpoint 7: ω, k, E are scaffolding, dismantled after the derivation; what remains is $\hbar, m$

The final equation has no ω, k, E , only $\hbar$ and m . Why: ω, k are properties of **one particular plane wave**, but the equation must hold for **all** plane waves, so it can't carry a specific ω, k . They appear mid-derivation and get eliminated by “derivative substitution” — “ ω ” $\leftrightarrow$ “ ∂_t acting,” “ k^2 ” $\leftrightarrow$ “ ∂_x^2 acting.” What remains, $\hbar$ (from de Broglie) and m (the particle's mass, from $E = \frac{p^2}{2m}$), are the universal constants.

4.1 One honesty: we did not use the $E\psi$ operator formalism

Textbooks often first write the “energy operator” $E \rightarrow i\hbar\partial_t$ and momentum operator $p \rightarrow -i\hbar\partial_x$, then operatorize $E\psi = \frac{p^2}{2m}\psi$ to get the equation. That formalism is useful when **adding a potential and generalizing to arbitrary states** (§7 uses its viewpoint), but for this pure free-particle step it introduces an intermediary $E\psi$ and instead creates notational confusion: $i\hbar\partial_t\psi$ already contains one ψ (being differentiated) on the left, and $E\psi$ writes another ψ on the right, looking as if “both sides were multiplied by an extra ψ .”

The route above — **directly substituting from the dispersion relation + derivatives** — is shorter and more honest: the left $\partial_t\psi$ contains one ψ , the right $\partial_x^2\psi$ contains one ψ , **one each, no more no less**, with no spare intermediary. Take this route.

5. Adding the potential $V(x)$

A free particle uses $E = \frac{p^2}{2m}$; a real particle is usually in some potential field (an electron pulled by the nucleus, a particle in a well), where the total energy gains a **potential** term:

$$E = \underbrace{\frac{p^2}{2m}}_{\text{kinetic}} + \underbrace{V(x)}_{\text{potential}}$$

The potential $V(x)$ is the energy a particle has by virtue of its **position** — set by what force field it’s in and where, independent of how fast it moves (gravity mgh , spring $\frac{1}{2}kx^2$, Coulomb $-\frac{e^2}{4\pi\epsilon_0 r}$); its relation to force is $F = -\frac{dV}{dx}$.

Checkpoint 8: $V = V(x)$ is a function of position, independent of ψ ; $V\psi$ is a pointwise product, a genuine multiplication

V is **not** a function of ψ — it’s “the potential value at each position x ,” a definite number given the position, independent of the ψ to be solved (V is a given external field). The notation $V\psi$ is a **pointwise product** $[V\psi](x) = V(x) \cdot \psi(x)$: two functions of x multiplied to a new function of x , legitimate and standard. Unlike §4.1’s $E\psi$, it has no trap — because $V(x)$ is an ordinary function (no derivative, no ψ), and as an operator it’s the honest “multiply by $V(x)$.”

Multiply the potential-inclusive energy relation $\hbar\omega = \frac{\hbar^2 k^2}{2m} + V$ by ψ and run the same substitution — terms with ω, k become derivatives, V (no ω, k) stays as is:

$$i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\partial_x^2\psi + V(x)\psi$$

So V is brought in by “adding a term to the energy relation”: the same “energy relation $\rightarrow$ derivative substitution” machine — whatever you add to the energy, the RHS gains. $\frac{p^2}{2m}$ has $p \rightarrow$ becomes the second derivative $-\frac{\hbar^2}{2m}\partial_x^2$; $V(x)$ has no $p \rightarrow$ stays as a multiplication on ψ .

6. Generalizing to three dimensions

Replace the 1D spatial derivative ∂_x^2 with the 3D Laplacian $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$ (each direction repeats the 1D derivation; momentum $p^2 = p_x^2 + p_y^2 + p_z^2$ corresponds to the sum of three second derivatives):

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\mathbf{r})\psi$$

This is the complete (non-relativistic, single-particle) Schrödinger equation. **1D $\rightarrow$ 3D is a technical generalization, not hard**; the real hurdle is the “describe a particle with a wave” seam in §2 — that’s where the post’s weight lies.

7. A few checkpoints that settled out

7.1 Why you can’t cancel the ψ on both sides

ψ is **not a common factor multiplied on both sides**; it’s the object **acted on** by an operator:

$$\underbrace{i\hbar \partial_t}_{\text{operator}} \psi = \left(\underbrace{-\frac{\hbar^2}{2m} \partial_x^2 + V(x)}_{\text{operator}} \right) \psi$$

“Acting” isn’t “multiplying”: $\partial_t \psi$ is “differentiate ψ ,” not “the number ∂_t times ψ .” Analogy: $\frac{d}{dx} f = 3f$ — you can’t cancel f to get $\frac{d}{dx} = 3$ (meaningless). And the equation’s **entire content** is precisely “the left and right, **different** operators, give the same result **when acting on the same ψ** ” — a **constraint** on ψ (only specific ψ satisfy it). Cancel ψ and the constraint vanishes, and you’d be claiming “two different operators are equal,” which is meaningless.

ψ appears three times with different statuses: in $\partial_t \psi$ and $\partial_x^2 \psi$ it is **differentiated**, inseparable; in $V\psi$ it is a **multiplicative factor**. Three different kinds — you can’t cancel them together as a common factor.

7.2 Why ψ must be complex (the aesthetic leap from 04 to 05)

04’s string ψ is **real** (displacement is an actual height, sin suffices, e^{ikx} is just mathematical convenience, take the real part at the end). 05’s quantum ψ is **essentially complex** — taking the real part would be wrong. The reason is this very equation: it’s **first order in t plus an i** . Look at the time part of the plane wave, $e^{-i\omega t}$:

- It’s a **pure phase rotation** — $|\psi|^2 = |e^{-i\omega t}|^2 = 1$ is time-independent (**probability conserved!**), while the phase rotates in the complex plane.
- If ψ were real (only sin or cos), a first-order time derivative can’t produce this “modulus-fixed, phase-rotating” structure, and probability wouldn’t be conserved.

So ψ is complex for **probability conservation + phase evolution**. This is the aesthetic leap from 04 to 05: **04’s wave swings up and down on the real axis (displacement); 05’s wave rotates in the complex plane (phase), modulus unchanged (probability conserved)**. In 05, e^{ikx} isn’t “a convenient packaging of sin” — the wave really lives in the complex plane and rotates.

7.3 It has no direct pictorial meaning like $F = ma$

$F = ma$ is pictorial because force and acceleration are visible realities. The Schrödinger equation operates on a **complex probability amplitude** ψ , for which the brain has no intuition, so reading no “picture” out of it is normal — not a failure of comprehension.

What you can grasp is a **structural image**: read term by term,

$$\underbrace{i\hbar\partial_t\psi}_{\text{total energy acting}} = -\underbrace{\frac{\hbar^2}{2m}\nabla^2\psi}_{\text{kinetic}} + \underbrace{V\psi}_{\text{potential}}$$

— “**how the wave function evolves in time**” is set by “**its total energy (kinetic + potential)**,” i.e. $E = \frac{p^2}{2m} + V$ turned into operators. The closest-to-pictorial analogy: $-\nabla^2\psi$ is “curvature” (how sharply the wave bends), so the equation roughly says “**the wave evolves/flows faster where it bends more sharply (large kinetic energy)**,” with ψ like something flowing in the complex plane, curvature (kinetic) and height (potential) together driving the flow. But this is still an analogy — its meaning is mainly in **being computable and predictive** (energy levels, tunneling, interference), not in a picture you grasp at a glance. In this it differs in nature from Newton’s equation; don’t expect the $F = ma$ kind of intuition.

8. This block’s IOU ledger

IOU	Content	When settled
Planck’s constant h	where the h in de Broglie comes from	Settled: Part 2, <i>Squeezing Planck’s Constant Out of Blackbody Radiation</i> (forces $\varepsilon = h\nu$, but h ’s value is a fit; all that’s derivable is “it must exist”)
de Broglie postulate	why a particle is a wave	Ch. 09 QFT (there particles are field excitations, wave nature is the field’s nature; $p = \hbar k$ is a consequence of the field’s translation symmetry — at the cost of a new set of postulates; this moves the IOU one layer deeper, not closes it)
Born rule	$ \psi ^2 = \text{probability}$	postulate, not derivable (accepted in 03); math only reaches “normalized nonnegative weights”

IOU	Content	When settled
$E = \frac{p^2}{2m}$	using non-relativistic energy	settled by Ch. 08 relativistic quantum equations (Schrödinger can't handle photons precisely because it uses non-relativistic E ; a scope issue)
why this E - p relation	the independent third input	decided by “what kind of particle”; the relativistic relation $\rightarrow$ Klein–Gordon/Dirac (08)
the form of $V(x)$	what it actually is	given by the specific physical system, not derived (the equation only prescribes how V enters, not what V is)

These IOUs don't damage first principles — first principles is “knowing why each step holds and honestly marking the black boxes,” not welding every foundation slab shut. Especially here, **the two deepest IOUs ($\hbar$, de Broglie) are postulate-level by nature; being underivable is their nature, not our oversight.**

9. Isomorphism and divergence with 04, and the interface

Isomorphism (similar): 04 and 05 share a whole **mathematical structure of waves** — eigenfunction bases, plane waves, dispersion relations, the “equation $\leftrightarrow$ dispersion” machine, mode decomposition. This layer is the same skeleton.

Divergence (two different scenarios):

	04 1D string	05 quantum particle
what ψ is the bottom derivation direction	real displacement medium + Newtonian mechanics equation $\rightarrow$ dispersion	complex probability amplitude no medium; a fundamental law dispersion $\rightarrow$ equation (no underlying equation; reverse-engineered from constraints)
dispersion relation boundary / basis time derivative	$\omega = ck$ (linear, dispersionless) fixed $\rightarrow \sin(k_n x)$, discrete second order (state = position + velocity)	$\omega \propto k^2$ (quadratic, dispersive) free $\rightarrow e^{ikx}$, continuous first order (state = ψ)
where the wave oscillates	swings on the real axis	rotates in the complex plane (modulus fixed = probability conserved)

So it's **isomorphic but not the same**: the mathematical skeleton is similar, the physical content diverges. No need to force them into one thing.

Interface (what's next):

- The “wave function + superposition + complex phase” here $\rightarrow$ 06 (states, operators, measurement, spin) upgrades it into a **Hilbert-space state vector** and makes the operator formalism $E \rightarrow i\hbar\partial_t, p \rightarrow -i\hbar\partial_x$ explicit.
- The IOU left by using non-relativistic $E = \frac{p^2}{2m} \rightarrow$ 08 (relativistic quantum equations) swaps the dispersion for the relativistic $E^2 = (pc)^2 + (mc^2)^2$, giving Klein–Gordon / Dirac.
- The pattern “swap in a different dispersion relation to get a different wave equation” is first genuinely cashed out in 05 (string $\omega = ck \rightarrow$ quantum $\omega \propto k^2$); every later physical system swaps the dispersion once more.